

Tucson traffic estimate

Sunday 1:00 - 3:00 AM — instantaneous vehicle count

Working estimate for investigative context. This document distinguishes between data points grounded in published sources and figures derived from structured estimation. All estimated figures should be cross-referenced with local data before use in any published analysis.

Metro baseline

Registered vehicles [SOURCED]	Road network [SOURCED]	Daily VMT [ESTIMATED]
~700,000	~4,800 lane-mi	20-25M
Census/ACS + ADOT	City + county + interstates	Vehicle-miles traveled

Sunday 1-3 AM snapshot

National and regional traffic count data consistently shows this window as the lowest-traffic period of the week [SOURCED]. FHWA and state DOT continuous count stations report it carries roughly 1-2% of daily traffic volume and 5-8% of peak-hour volume.

Central estimate: 5,000 - 6,000 vehicles
on all Tucson-area roads at any given moment

Range: 4,000 - 8,000 vehicles. Could be off by 30-40% in either direction. Based on national hourly distribution factors applied to Tucson VMT and road network size. No Tucson-specific study identified for this exact figure. [ESTIMATED]

Peak-hour vehicles	Sun 1-3 AM vs peak [SOURCED]	% of daily volume [SOURCED]
80-100K	5-8%	1-2%
Weekday reference	FHWA hourly distribution	Weekly traffic low point

Breakdown by road type [ESTIMATED]

Pattern-based reasoning, not Tucson-specific count data. This is the roughest layer of the estimate.

Road type	Est. vehicles	Notes
I-10 / I-19 corridors	500 - 1,000	Long-haul trucking runs 24/7; bar-close traffic feeds onto I-10 ramps
Major arterials (Speedway, Broadway, Grant, Oracle, etc.)	1,500 - 3,000	Spread across dozens of corridors; rideshare surge around 2 AM bar close
Collectors + local streets	2,000 - 4,000	Bar/restaurant close-out, shift workers (hospitals, hotels, warehouses), convenience ru

Who is on the road at this hour

- **Bar / restaurant close-out** — AZ bars serve until 2 AM — regulatory fact that shapes the late-night traffic curve
- **Rideshare + taxi drivers** — Surge pricing window around bar close; concentrated near entertainment districts
- **Shift workers** — Hospitals, hotels, warehouses, distribution centers — predictable patterns
- **Long-haul truckers** — Interstate corridors operate 24/7 regardless of day of week
- **Late-night errands** — Convenience store runs, fast food, pharmacy — scattered and unpredictable
- **Unaccounted movement** — People who are simply driving — no destination category

Corridor-specific: Catalina Foothills area

Applied to the Nancy Guthrie case geography. Starting point: 5820 N Camino Escalante, Tucson AZ (Catalina Foothills). The map tool calculates drive-time radii, camera coverage, and escape route scenarios from this location.

Key egress corridors from residence

Corridor	Direction	To I-10 access	Camera density	Late-night character
Swan Rd south	S	~6 mi / 10 min	Low - residential	Residential feeder; minimal surveillance; connects to I-10 @ Palo Verde / Countr
Campbell Ave south	S-SW	~5 mi / 9 min	Low-moderate	Residential to commercial transition; hits I-10 @ Congress/Speedway area
Tanque Verde / Catalina Hwy	NE	Away from I-10	Very low	Dead-ends into Coronado NF / Mt. Lemmon; no through-traffic at 1-3 AM
Sunrise Dr west to Oracle Rd	W	~7 mi / 12 min	Low	Connects to Oracle (SR-77) north toward Oro Valley; or south to I-10
Kolb Rd south	SE	~8 mi / 13 min	Low-moderate	Long residential stretch; connects to I-10 east toward Houghton/Vail

All corridor distances, travel times, and camera density assessments are [ESTIMATED] based on map tool analysis. Camera data combines ADOT highway cameras (verified positions) and OSM community-mapped cameras (variable completeness).

I-10 corridor — ADOT camera coverage

The map tool catalogs ADOT highway cameras along I-10 from Cortaro Rd (Exit 246) through Houghton Rd (Exit 275). Camera spacing averages approximately 1-2 miles through the urban core but thins significantly past the Valencia Rd interchange heading southeast and past Cortaro heading northwest. The WB I-10 escape corridor analysis in the map tool identifies this thinning as a vulnerability — once past Marana, camera gaps widen to 5+ miles.

What this means for the investigation

At 1-3 AM on a Sunday, a vehicle departing the Catalina Foothills area would encounter very sparse traffic on the local arterials (Sunrise Dr, Swan Rd, Campbell Ave). The residential character of these corridors means few if any commercial surveillance cameras. Once on I-10, ADOT cameras provide periodic coverage through the urban core, but gaps exist and widen outside the metro. The total number of potential witness vehicles on any single corridor at that hour is likely in the low hundreds at most — possibly fewer than 100 on residential feeder roads.

Law enforcement reviewing this window faces a needle-in-haystack problem: enough vehicles on the road that movement would not be conspicuous, but few enough that identifying a specific vehicle requires targeted data sources (ALPR, gas station cameras, cell tower records) rather than volume-based pattern matching.

Data confidence summary

Data point	Confidence	Source / basis
Metro population + registered vehicles	SOURCED	Census/ACS, ADOT registrations
Sunday 1-3 AM = weekly low	SOURCED	FHWA, state DOT continuous count stations
1-2% of daily / 5-8% of peak ratios	SOURCED	Highway Capacity Manual, FHWA Traffic Monitoring Guide
AZ bar closing time (2 AM)	SOURCED	ARS 4-244 regulatory fact

Instantaneous vehicle count (5-6K)	ESTIMATED	National factors applied to Tucson VMT; no local study; +/- 30-40%
Road-type breakdown	ESTIMATED	Pattern reasoning only; no Tucson-specific count data
Corridor distances / travel times	ESTIMATED	Map tool routing; actual times vary with conditions
Camera density assessments	ESTIMATED	ADOT verified + OSM community-mapped; private cameras not included

Cross-reference: sources for harder numbers

- **Pima County + City of Tucson permanent traffic count stations**
Hourly volumes available through ADOT Traffic Data portal. Would replace estimated instantaneous count with observed data.
- **TPD / PCSD calls for service and traffic stop logs**
Calls for service, traffic stops, DUI enforcement activity during 1-3 AM window. Indirect but concrete proxy for vehicles officers are encountering.
- **ADOT traffic count maps — corridor-specific**
Some stations report hourly-level data for individual corridors. Could validate or replace the road-type breakdown.
- **Pima Association of Governments (PAG) regional traffic model**
May have time-of-day VMT distributions specific to Tucson metro rather than relying on national factors.
- **ALPR data (if accessible via FOIA)**
Automated License Plate Reader logs from TPD or PCSD would give actual vehicle counts at specific locations during this window.

This document is a working analytical estimate prepared for investigative context. Figures tagged [ESTIMATED] have not been verified against local data and should not be cited as fact. Generated by Claude (Anthropic) at the request of Subtxt Press.